



# Transient PDE-constrained optimization for hyperbolic phenomena

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Design geometry · Inverse identification problems  
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# Transient optimization

## Objective functional

$$\mathcal{J}(\rho, u, v) = \int_0^T \int_{\Omega} \mathcal{J}_{\Omega}(t, \rho, u(t), v(t)) dx dt + \mathcal{J}_T(\rho, u(T), v(T)).$$

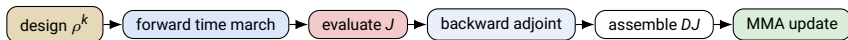
## PDE-constrained minimization

$$\begin{aligned} \min_{\rho \in \mathcal{Q}} \quad & \mathcal{J}(\rho, u), \\ \text{s.t.} \quad & \mathcal{R}(\rho, u; w) = 0 \quad \forall w \in \mathcal{U}, \\ & u \in \mathcal{U}. \end{aligned}$$

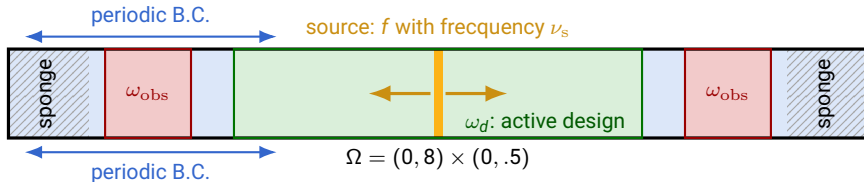
## Elastodynamics PDE

$$\begin{aligned} \rho_{\rho} \ddot{u} + C_{\rho} \dot{u} - \nabla \cdot \sigma(\rho, u) &= f & \Omega \times (0, T), \\ \sigma \cdot n + \eta \dot{u} &= g & \partial\Omega \times (0, T), \\ u(0, x) &= u_0, \quad \dot{u}(0, x) = v_0. \end{aligned}$$

## Adjoint based minimization.



# Band waveguide: ask the optimizer to build a reflector



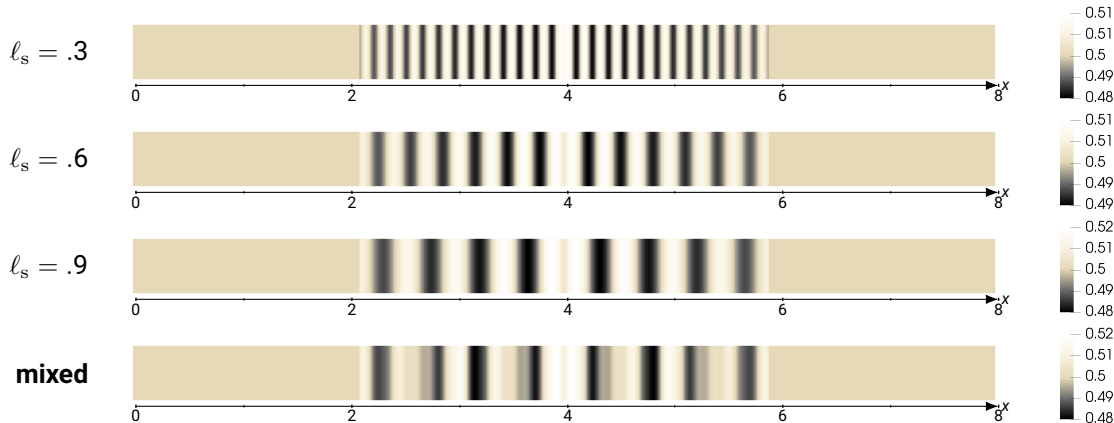
## Objective Function

$$\min_{0 \leq \rho_h \leq 1} \int_0^6 \int_{\omega_{\text{obs}}} |u|^2 \quad \text{s.t.} \quad \int_{\omega_d} \rho_h = .5 |\omega_d|$$

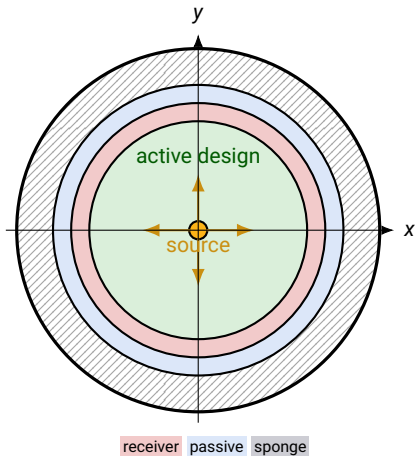
## Parameters

- Lagrange  $P_2$  state variable (409,920 DoFs).
- DG  $P_0$  design variable  $\rightarrow$  Helmholtz filter.
- Source wavelength  $\ell_s$ : .3, .6, .9 and a mixed carrier.

# Band result: the forcing wavelength selects the layer scale



# Spherical waveguide: radial 3D setting



Suppress outgoing energy

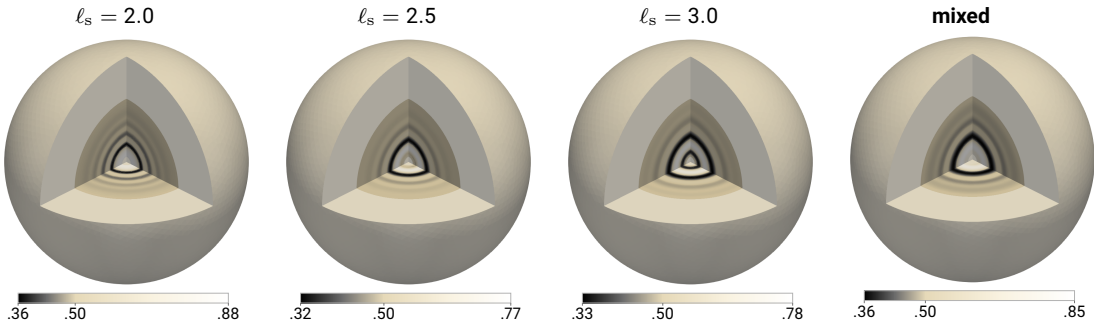
$$\min_{0 \leq \rho_h \leq 1} \mathcal{J} = \int_0^9 \int_{6 < r < 7} |u|^2,$$

$$\text{s.t.} \quad \int_{.5 < r < 6} \rho_h = \frac{1}{2} |\{.5 < r < 6\}|.$$

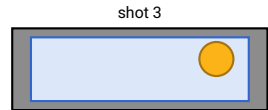
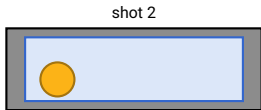
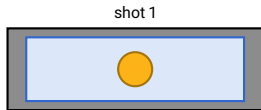
## Parameters

- DG  $P_0$  design variable  $\rightarrow$  Helmholtz filter.
- Mesh: 2,017,520 tetrahedra.
- Lagrange  $P_2$  state variable (16,308,294 DoFs).
- 1,800 RK4 steps.
- $\ell_s$ : 2.0, 2.5, 3.0 and a mixed carrier.

# Spherical waveguide: optimized designs



# Inverse problem: illuminate three times, observe elsewhere



## Objective function

$$J(\rho) = \frac{1}{2} \sum_{k=1}^3 \int_0^T \|u_k(\rho) - d_k\|_{\Gamma_{\text{obs}}^{(k)}}^2 dt$$

- $\Gamma_{\text{obs}}^{(k)}$ : load-free observed surface.
- $\Gamma_{\text{src}}^{(k)}$ : Active load.
- Black region: damping collar.
- $\Gamma_{\text{obs}}^{(k)} \cap \Gamma_{\text{src}}^{(k)} = \emptyset$

## Reference data (per shot)

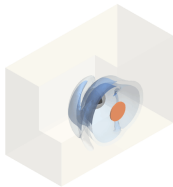
- Mesh: 2.05M hexahedra.
- Reference design  $\rho_{\dagger}$  DG space,  $P_0$ : (2.05M DoFs).
- State variables  $(u, v)$  Lagrange,  $P_3$ : (335M DoFs).
- 25,000 time steps.

## Production / optimization

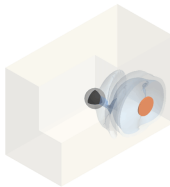
- Mesh: 2.05M hexahedra.
- Design  $\rho_h$  DG space  $P_0$ : (2.05M DoFs); 365K active.
- State  $(u, v)$  Lagrange  $P_2$ : (100M DoFs).
- 6,250 symplectic-euler steps.

# Three-source forward wave propagation

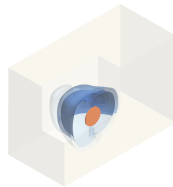
**Shot 1: central source**



**Shot 2: left corner**

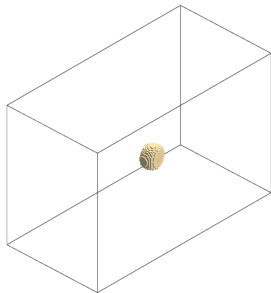


**Shot 3: right corner**



# Three-source inverse identification: results

Recovered  $\rho^{26}$ , raw  $\rho < 0.7$



Prescribed reference density,  $\rho_{\dagger}$

